Yesterday 10:43 PM

You said:

PDF **piecuch2005.pdf**
PDF

Can you look into the method CR-CC(2,3) and determine how to compute analytical gradients of the delta(2,3) correction? I have uploaded article `piecuch2025.pdf` , which is the authoritative reference on this method. You can also access the CCpy code on GitHub https://github.com/piecuch-group/ccpy/tree/main/ccpy to see the detailed Python implementation.

We need to extend typical response/gradient CC frameworks (e.g., similar to what was done for CCSD(T)) to CR-CC(2,3)

Worked for 4m 12s >

ChatGPT said:

Yes. CR-CC(2,3) admits an analytic-gradient formulation using a constrained CC Lagrangian. The main additions to a conventional CCSD(T) framework are:

Outputs

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PDF piecuch2005.pdf

- Response of the **CCSD Λ amplitudes**, because they explicitly enter the correction.
- Derivatives of the **amplitude-dependent triples denominators**, including their three-body contributions for full CR-CC(2,3).
- Orbital-response terms consistent with the orbital convention used to evaluate those denominators.

The closest existing framework is **ΛCCSD(T)**. Taube and Bartlett explicitly developed its stationary formulation and analytic derivatives: *J. Chem. Phys.* **128**, 044111 (2008), DOI: 10.1063/1.2830237. That provides a more direct starting point than ordinary CCSD(T), although its triples expressions cannot simply be reused unchanged. pubmed.ncbi.nlm.nih.gov

One bibliographic clarification: the uploaded reference is Piecuch and Włoch's **2005** paper, *JCP* **123**, 224105. I used its Eqs. (26)–(31), checked against the rendered PDF:

Below is my derivation of the gradient framework and its mapping onto CCpy. It is not a claim that a production CR-CC(2,3) nuclear-gradient implementation has already been completed.

1. Write the correction as a function of the actual CCSD variables

Use independent singles/doubles excitation coordinates t_μ, λ_μ, with

$$T = T_1 + T_2, \qquad \Lambda = \Lambda_1 + \Lambda_2, \qquad \bar{H} = e^{-T}$$

Let K denote a unique triple excitation, $i < j < k,\ a < b < c$. Define

$$M_K(t) = \langle K|\bar{H}|0\rangle,$$

$$N_K(t,\lambda) = \langle 0|\Lambda\bar{H}|K\rangle,$$

$$D_K(t) = E_{\mathrm{CCSD}} - \langle K|\bar{H}|K\rangle.$$

Then full CR-CC(2,3), corresponding to CCpy's **D variant**, is

$$\boxed{E_{\mathrm{CR}} = E_{\mathrm{CCSD}} + \delta, \qquad \delta = \sum_K \frac{N_K M_K}{D_K}.}$$

With unrestricted antisymmetric spin-orbital summation, replace $\sum_K$ by $\frac{1}{36}\sum_{ijkabc}$.

The denominator variants share the same numerator:

Variant	Denominator
A	Bare Fock orbital-energy differences
B	One-body $\bar{H}$ contributions
C	One- and two-body $\bar{H}$ contributions
D	One-, two-, and three-body $\bar{H}$ contributions

The full method in the uploaded paper is D; A is equivalent to $\mathrm{CCSD}(2)_T$ with canonical HF orbitals. truhlar.chem.umn.edu

2. Differentiate all three factors

For any parameter q,

$$\boxed{\delta_q = \sum_K \left[\frac{N_K}{D_K}(M_K)_q + \frac{M_K}{D_K}(N_K)_q - \frac{N_K M_K}{D_K^2}(D_K)_q\right]}$$

It is useful to introduce algebraic triples quantities

$$r_K = \frac{M_K}{D_K}, \qquad \ell_K = \frac{N_K}{D_K}.$$

Thus,

$$\boxed{\delta_q = \sum_K [\ell_K (M_K)_q + r_K (N_K)_q - \ell_K r_K (D_K)_q}$$

These are also the reverse-mode differentiation seeds:

$$\frac{\partial \delta}{\partial M_K} = \ell_K, \qquad \frac{\partial \delta}{\partial N_K} = r_K, \qquad \frac{\partial \delta}{\partial D_K} = -$$

Here r_K is an auxiliary quantity, **not a converged CCSDT amplitude**.

For a nuclear displacement x, however, Equation (1) alone does not finish the gradient:

$$\frac{d\delta}{dx} = \delta_x + \delta_t^T t^x + \delta_\lambda^T \lambda^x + \text{orbital/basis respor}$$

The goal is to eliminate the explicit t^x, λ^x calculations for every coordinate.

3. Constrain both the right- and left-CCSD equations

Define the CCSD residual and Lagrangian,

$$R_\mu(t) = \langle \mu | \bar{H} | 0 \rangle = 0,$$

$$\mathcal{L}_0(t, \lambda) = E_{\text{CCSD}}(t) + \lambda^T R(t).$$

Write

$$A_{\mu\nu} = \frac{\partial R_\mu}{\partial t_\nu}, \qquad g_\mu(t, \lambda) = \frac{\partial \mathcal{L}_0}{\partial t_\mu}.$$

The ordinary CCSD Λ equations are

$$g = E_t + A^T \lambda = 0.$$

Now construct

$$\boxed{\mathcal{L}_{\mathrm{CR}} = \mathcal{L}_0 + \delta + z^T R + w^T g.} \qquad (3)$$

At converged CCSD and left-CCSD solutions, $R = g = 0$, so this functional equals E_{CR}.

The two new multipliers have distinct jobs:

- w removes the response of the **original CCSD** Λ.
- z removes the remaining response of T.

This is the same structural reason that the ΛCCSD(T) stationary formulation introduces two additional conjugate variables. Taube's dissertation gives that construction explicitly in Sec. 3.2.2.
ufdcimages.uflib.ufl.edu

Let

$$s_t = \delta_t, \qquad s_\lambda = \delta_\lambda,$$

and define the CCSD Lagrangian Hessian with respect to amplitudes,

$$B_{\mu\nu} = \frac{\partial g_\mu}{\partial t_\nu} = \frac{\partial^2 \mathcal{L}_0}{\partial t_\mu \partial t_\nu}.$$

Stationarity of Equation (3) with respect to λ gives

$$\boxed{Aw = -s_\lambda.} \qquad (4)$$

Stationarity with respect to t, evaluated where $g = 0$, gives

$$\boxed{A^T z = -s_t - B^T w.} \qquad (5)$$

For independent real coordinates, B is symmetric, but retaining the transpose makes the differentiation

convention explicit.

These are two sequential CCSD-sized linear solves. Neither requires iterative triples. Moreover, B need not be constructed: only the product $B^T w$ is required.

A useful check is ordinary CCSD(T). Its correction has no independent CCSD-Λ dependence, so $s_\lambda = 0, w = 0$, and Equation (5) becomes the familiar correction to the CCSD multiplier equation.

4. The analytic derivative of the correction

At fixed orbital coordinates, let a subscript x mean an explicit derivative holding t, λ fixed. Stationarity gives

$$\frac{dE_{\mathrm{CR}}}{dx} = E_x + (\lambda + z)^T R_x + \delta_x + w^T g_x.$$

Since

$$\frac{dE_{\mathrm{CCSD}}}{dx} = E_x + \lambda^T R_x,$$

the requested correction derivative is

$$\boxed{\frac{d\delta}{dx} = \delta_x + z^T R_x + w^T g_x.} \qquad (6)$$

Orbital and basis response must subsequently be included.

Equation (6), together with Equations (2), (4), and (5), is the central implementation result. **A modified left-CCSD solve alone is insufficient:** it omits the w equation and associated terms.

For explicit source construction, because excitation operators commute,

$$\frac{\partial \bar{H}}{\partial t_\mu} = [\bar{H}, \tau_\mu].$$

Consequently,

$$(M_K)_{t_\mu} = \langle K|[\bar{H}, \tau_\mu]|0\rangle,$$

$$(N_K)_{t_\mu} = \langle 0|\Lambda[\bar{H}, \tau_\mu]|K\rangle,$$

$$(D_K)_{t_\mu} = \langle 0|[\bar{H}, \tau_\mu]|0\rangle - \langle K|[\bar{H}, \tau_\mu]|K\rangle.$$

The left-amplitude source is particularly simple:

$$\boxed{(s_\lambda)_\mu = \sum_K r_K \langle \mu|\bar{H}|K\rangle.} \tag{7}$$

For the $B^T w$ contribution, define $W = \sum_\mu w_\mu \tau_\mu$.
Then

$$\boxed{(B^T w)_\nu = \langle 0|(1+\Lambda)[[\bar{H}, \tau_\nu], W]|0\rangle.} \tag{8}$$

These are formal expressions for deriving or validating factorized contractions—not instructions to build determinant-space matrices in production.

5. A stationary triples functional makes the denominator terms explicit

Instead of differentiating the quotient directly, introduce independent r_K, ℓ_K and use

$$\mathcal{F}_3 = \sum_K \left[\ell_K M_K + N_K r_K - \ell_K D_K r_K\right].$$

Stationarity yields

$$D_K r_K = M_K, \qquad D_K \ell_K = N_K,$$

and $\mathcal{F}_3 = \delta$. Therefore an equivalent complete amplitude Lagrangian is

$$\boxed{\mathcal{L}_{\mathrm{CR}} = \mathcal{L}_0 + z^T R + w^T g + \mathcal{F}_3.} \qquad (9)$$

This is convenient for symbolic derivation, automatic differentiation, and density construction. In particular, the term

$$-\ell_K r_K (D_K)_q$$

cannot accidentally disappear.

For variant D,

$$D_K = -\langle K|(\bar{H}_1 + \bar{H}_2 + \bar{H}_3)|K\rangle.$$

The scalar part of $\bar{H}$ has already canceled against E_{CCSD}. Thus one should either differentiate $E_{\mathrm{CCSD}} - \bar{H}_{KK}$ consistently, or differentiate the scalar-canceled expression. Adding a separate E^x_{CCSD} term to CCpy's already scalar-canceled denominator would double count it.

Also, the $\bar{H}_3$ dependence does **not** imply a need for three-electron integral derivatives. For an ordinary electronic Hamiltonian, those effective three-body terms are built from the underlying one-/two-electron integrals and cluster amplitudes.

6. Mapping this onto CCpy

I inspected repository commit `cf72da4865e054f7a935e2df27e920cf5e335435`. The relevant components are:

Component	Existing code	Gradient extension
Correction driver	moments/crcc23.py	Accumulate amplitude and integral derivati
Triples contractions	lib/fortran/crcc_loops.f90	Reverse contra using $\ell, r, -\ell r$

Component	Existing code	Gradient extension
and denominators		
Three-body diagonal	hbar/diagonal.py	Differentiate bo integrals and T
Similarity transformation	hbar/hbar_ccsd.py	Propagate derivatives bac through $\bar{H}(T,$
Right Jacobian action	eomcc/eomccsd.py	Reuse for the inhomogeneou solve
Left Jacobian action	left/left_ccsd.py	Reuse for $A^T z$ construct $B^T w$

In the same-spin `crcc23A_opt` kernel—the routine name denotes a **spin case**, not denominator variant A—the quantities are directly identifiable:

Plain text

```
M_K = temp1
N_K = temp2 + temp3
LM  = M_K * N_K
deltaD += LM / D
```

Therefore, a reverse pass starts with

Plain text

```
adj_M = N_K / D
adj_N = M_K / D
adj_D = -M_K * N_K / D**2
```

For example, `aaa_H3_aaa_diagonal` contains

$$d^O_{aij} = \sum_e v^{ae}_{ij} t^{ae}_{ij}, \qquad d^V_{aib} = -\sum_m v^{ab}_{im} t^{ab}_{im}.$$

Its derivative includes both factors:

$$(d^O_{aij})_q = \sum_e \left[(v^{ae}_{ij})_q \, t^{ae}_{ij} + v^{ae}_{ij} \, (t^{ae}_{ij})_q \right].$$

The first contribution feeds integral derivatives/densities; the second feeds s_t.

Two implementation details deserve particular care:

- **Differentiate through $\bar{H}$.** Treating the `H` argument of `calc_crcc23` as independent of T misses a large part of s_t.
- **Use the correct excitation-space metric.** The equations above use independent coordinates. CCpy's full antisymmetric spin blocks require consistent permutation weights; a plain Euclidean transpose of flattened redundant arrays is not automatically the physical adjoint.

For automatic differentiation, differentiate the **raw residual equations and contractions**, not the DIIS iterations or preconditioned updates. Also ensure that any residual identities valid only at converged T are handled consistently when constructing derivatives away from that solution.

7. Completing the molecular nuclear gradient

Equations (4)–(6) eliminate amplitude response. A nuclear gradient additionally needs the usual SCF orbital constraints and AO overlap terms.

Schematically, augment the Lagrangian by

$$\mathcal{L}_{\text{full}} = \mathcal{L}_{\text{CR}} + Z^T_{\text{orb}} f_{\text{constraints}} - \text{Tr}\big[X(C^T S C -$$

Solve the orbital adjoint equations, then form relaxed one- and two-particle derivative densities. With spin-orbital antisymmetrized integrals and consistent density normalization,

$$E^x_{\mathrm{CR}} = E^x_{\mathrm{nuc}} + \sum_{pq} \gamma_{pq} h^x_{pq} + \frac{1}{4} \sum_{pqrs} \Gamma_{pqrs} \langle pq \| rs \rangle^x + \sum_{pq} \lambda^{\mathrm{orb}}_{pq} S^x_{pq} \tag{10}$$

The correction-only gradient follows from the differences between CR and CCSD relaxed densities and overlap multipliers.

A conventional CCSD(T) gradient implementation already supplies much of the machinery for integral derivatives, density contraction, orbital relaxation, and overlap contributions. Hald *et al.*'s Lagrangian implementation is a useful architectural reference.
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One extra issue is orbital convention. A determinant-diagonal approximation to the triples resolvent is generally not invariant under arbitrary occupied–occupied or virtual–virtual rotations. For B–D, one must not automatically discard these responses using CCSD's orbital-invariance argument.

For a canonical-HF implementation, differentiate the canonicalization conditions as well as the occupied–virtual SCF conditions, or demonstrate an equivalent treatment in the chosen gradient formalism. Frozen-core boundaries and degenerate orbital subspaces need consistent handling.

8. What I verified, and how I would implement it

I tested Equations (4)–(8) using a three-electron/six-spin-orbital model with a real two-body Hamiltonian

$$H(x) = H_0 + xV.$$

I evaluated the full determinant-space similarity transform, solved CCSD and Λ, and compared the analytic correction derivative against finite differences with both solutions reconverged at each displacement:

Quantity	Result
Analytic $d\delta/dx$	$3.00087632010749 \times 10^{-4}$
Finite difference, $h = 10^{-4}$	$3.00087631361295 \times 10^{-4}$
Finite difference, $h = 10^{-5}$	$3.00087632000567 \times 10^{-4}$
Fixed-amplitude partial derivative only	$1.35983947738556 \times 10^{-4}$

The final discrepancy was approximately 10^{-14}. This checks the response algebra and signs; **it does not validate molecular orbital response or a spin-adapted CCpy implementation**.

My implementation sequence would be:

1. Build a differentiable, small-system correction evaluator matching CCpy's A–D energies.
2. Verify s_t, s_λ, and integral derivatives independently.
3. Implement the two adjoint solves using existing CCSD Jacobian actions.
4. Validate Equation (6) against reconverged finite differences of Hamiltonian parameters.
5. Add canonical orbital response, relaxed densities, and AO derivative contraction.
6. Validate molecular gradients, including stretched geometries, translation/rotation checks, and denominator-sensitive cases.
7. Stream triples blocks and reverse their contractions to avoid storing complete o^3v^3 arrays.

The expected factorized cost retains noniterative triples-scale work, plus CCSD-scale response iterations. The central development task is **a Λ-dependent triples-correction Lagrangian with complete denominator differentiation**, using

ΛCCSD(T) as the response template and CCpy's actual CR-CC(2,3) expressions as the energy definition.

Today 9:37 AM

You said:

Ok. Let's first start with a CCSD gradient. Assume I have the normal-ordered component of 1- and 2-RDMs as well as the gradient of the HF wave function computed. How do we proceed?

Worked for 41s >

ChatGPT said:

The next step is to add the correlation contribution to the HF energy gradient, including orbital relaxation and the correlation part of the Pulay term. Your normal-ordered RDMs already handle the CC amplitude response **provided they are the CCSD Lagrangian densities constructed from converged T and Λ.**

There is one distinction to make immediately:

- If you have the **HF energy gradient** dE_{HF}/dx, you still need orbital response.
- If you have the **HF wave-function derivative**, including the MO coefficient derivatives C^x, you can directly evaluate the CCSD gradient without an additional orbital Z-vector solve.

I'll describe that direct route first, then how to replace C^x with a single Z-vector calculation.

1. Fix the density convention

Use spin-orbital notation, with antisymmetrized integrals $v_{pqrs} = \langle pq||rs\rangle$:

$$H = E_{\mathrm{HF}} + \sum_{pq} f_{pq}\{a_p^\dagger a_q\} + \frac{1}{4}\sum_{pqrs} v_{pqrs}\{a_p^\dagger a_q^\dagger a$$

Here the scalar includes nuclear repulsion, and braces denote normal ordering relative to the HF determinant.

Define your normal-ordered densities as

$$\gamma_{pq}^N = \langle 0|(1+\Lambda)e^{-T}\{a_p^\dagger a_q\}e^T|0\rangle,$$

$$\Gamma_{pqrs}^N = \langle 0|(1+\Lambda)e^{-T}\{a_p^\dagger a_q^\dagger a_s a_r\}e^T|0\rangle.$$

Then the CCSD Lagrangian is

$$\mathcal{L}_{\mathrm{CCSD}} = E_{\mathrm{HF}} + \sum_{pq} f_{pq}\gamma_{pq}^N + \frac{1}{4}\sum_{pqrs} v_{pqrs}\Gamma_{pqrs}^N.$$

At the converged solution, it equals the CCSD energy and is stationary with respect to T, Λ.

"Normal-ordered 2-RDM" must mean the expectation of the normal-ordered operator above. It is not generally the same as the full 2-RDM minus the HF 2-RDM, or the two-body cumulant.

2. If you have the HF orbital derivatives, the gradient is a direct contraction

Let a dot denote the derivative of an integral **in the moving HF MO basis**, including both AO-integral and MO-coefficient derivatives. Then

$$\boxed{\frac{dE_{\mathrm{CCSD}}}{dx} = \frac{dE_{\mathrm{HF}}}{dx} + \sum_{pq}\gamma_{pq}^N \dot{f}_{pq} + \frac{1}{4}\sum_{pqrs}\Gamma_{pqrs}^N \langle \dot{p}q||rs\rangle + \text{(1)}}$$

You do **not** differentiate γ^N or Γ^N: their implicit T, Λ response cancels by Lagrangian stationarity.

But $\dot{f}, \dot{v}$ must include orbital response. Contracting the densities with bare AO-integral derivatives transformed using fixed C is insufficient.

To construct these derivatives, write

$$C^x = CU^x.$$

For real orbitals, differentiating $C^T S C = I$ gives

$$U^x + (U^x)^T = -\bar{S}^x, \qquad \bar{S}^x = C^T S^x_{\mathrm{AO}} C.$$

Thus

$$U^x = -\frac{1}{2}\bar{S}^x + \kappa^x, \qquad (\kappa^x)^T = -\kappa^x.$$

The symmetric part accounts for the changing AO metric; the antisymmetric part describes orbital rotations.

Define the fixed-coefficient AO derivative transformed to MOs,

$$h^{[x]}_{pq} = \sum_{\mu\nu} C_{\mu p} C_{\nu q} h^x_{\mu\nu}.$$

Then

$$\boxed{\dot{h}_{pq} = h^{[x]}_{pq} + \sum_t U^x_{tp} h_{tq} + \sum_t U^x_{tq} h_{pt}.} \tag{2}$$

Likewise,

$$\boxed{\begin{aligned}\dot{v}_{pqrs} = {} & v^{[x]}_{pqrs} \\ & + \sum_t \big[U^x_{tp} v_{tqrs} + U^x_{tq} v_{ptrs} \\ & \qquad + U^x_{tr} v_{pqts} + U^x_{ts} v_{pqrt}\big].\end{aligned}} \tag{3}$$

Finally, because

$$f_{pq} = h_{pq} + \sum_i v_{piqi},$$

with i running over occupied spin orbitals,

$$\boxed{\dot{f}_{pq} = \dot{h}_{pq} + \sum_i \dot{v}_{piqi}.} \qquad (4)$$

Equations (1)–(4) give a complete direct-response implementation under these conventions.

Notice that $\dot{f}$ includes the response of the reference occupied orbitals through the two occupied indices in v_{piqi}. It is not just the derivative of the one-electron Hamiltonian.

3. If you only have the HF energy gradient, solve orbital response

An HF energy gradient generally does not provide U^x. HF stationarity lets its own gradient avoid explicit occupied–virtual orbital response, whereas the CCSD correlation Lagrangian is not stationary with respect to the HF orbitals.

For a first implementation, solve CPHF for each nuclear coordinate, obtain U^x, and use Equation (1). That is a useful reference implementation.

For an efficient implementation, eliminate coordinate-specific CPHF solves using an orbital Z-vector. This orbital-constrained Lagrangian construction is part of the standard CC gradient framework. trygvehelgaker.no

Let

$$\mathcal{L}_{\mathrm{corr}} = \sum_{pq} f_{pq}\gamma^N_{pq} + \frac{1}{4}\sum_{pqrs} v_{pqrs}\Gamma^N_{pqrs}.$$

Define the correlation orbital gradient

$$G_{ai} = \left.\frac{\partial \mathcal{L}_{\mathrm{corr}}}{\partial \kappa_{ai}}\right|_{T,\Lambda}.$$

You can construct G by differentiating the integral transformations in Equations (2)–(3) with respect to orbital rotations, contracting with the fixed densities.

Write the differentiated HF orbital conditions as

$$K\kappa^x + b^x = 0,$$

where K is the CPHF orbital Hessian, and b^x includes the explicit integral and overlap perturbations.

The correlation gradient has the form

$$E^x_{\mathrm{corr}} = \tilde{\mathcal{L}}^x_{\mathrm{corr}} + G^T\kappa^x.$$

Here $\tilde{\mathcal{L}}^x_{\mathrm{corr}}$ contains the AO-integral derivatives and the metric contribution $U^x = -\bar{S}^x/2$, but excludes the unknown orbital rotations.

Solve once:

$$\boxed{K^T Z = -G.} \tag{5}$$

Then

$$G^T\kappa^x = Z^T b^x,$$

and

$$\boxed{E^x_{\mathrm{CCSD}} = E^x_{\mathrm{HF}} + \tilde{\mathcal{L}}^x_{\mathrm{corr}} + Z^T b^x.} \tag{6}$$

This orbital Z is distinct from the CCSD Λ amplitudes.

For all-electron CCSD with complete occupied and virtual spaces, internal occupied–occupied and virtual–virtual rotations are redundant. Frozen

orbitals require additional care at the boundaries between frozen and correlated spaces.

4. The practical sequence

Given your inputs, I would proceed as follows:

1. **Verify the density definitions.** Confirm they contain Λ, and check that
$$\sum f\gamma^N + \tfrac{1}{4}\sum v\Gamma^N = E_{\mathrm{corr}}$$
at converged CCSD.
2. **Obtain AO derivatives** h^x, v^x, S^x, including all nuclear-attraction and basis-center contributions.
3. **Use explicit CPHF orbital derivatives first.** Build $\dot{h}, \dot{v}, \dot{f}$, then evaluate Equation (1).
4. **Compare against finite differences of fully reconverged CCSD energies.**
5. **Replace coordinate-specific orbital response with Equation (5)** once the direct implementation agrees.

Adding the HF gradient does not remove the correlation Pulay contribution. In the direct route, that contribution is already contained in the overlap-dependent part of U^x; **do not add another correlation overlap term afterward**.